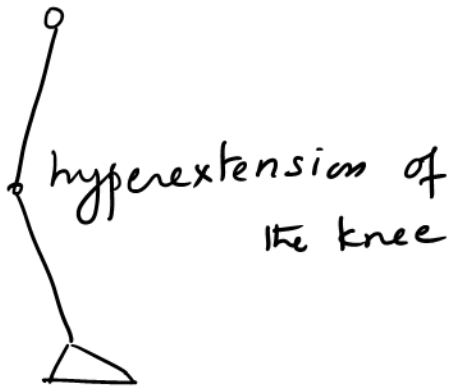
Excessive knee extension (EKE)



Normal stance phase knee flexion is lost

Hyperextension is seen typically in people with quadriceps paralysis following poliomyelitis

Hyperextension deformity called genu recurvatum

Joint frequently develops osteoarthritis later

EKE also caused by anatomical leg length discrepancy

Excessive knee flexion

Knee normally fully extended twice in the gait cycle – close to IC & mid-stance (close to HR)

Flexion contracture in the knee

Spasticity in the hamstrings would prevent full extension knee flexors

Knee may also flex excessively if normal plantarflexion during LR is prevented.

Inadequate dorsiflexion control

At IC - dorsiflexors prevent foot slap
Swing - dorsiflexors help the foot clear the ground by preventing toe drag.

Abnormal foot contact

Talipes calcaneus - deformity where the forefoot is pulled into extreme dorsiflexion (spasticity in ant. tibial muscles or weakness in triceps surae)
Weight is never taken by the forefoot

Result: Stance phase duration is ↓ (∵ no terminal forefoot stance)

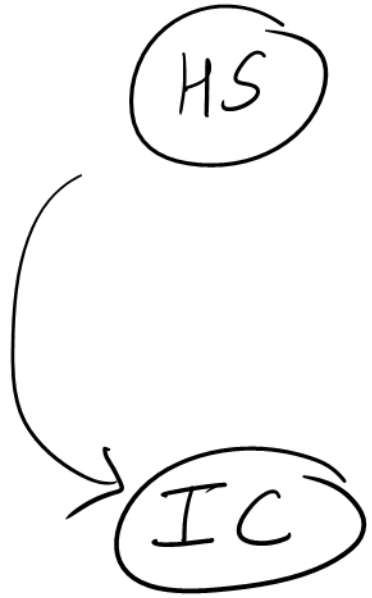
$\Rightarrow$ swing phase duration of the opp. side ↓

$\Rightarrow$ opp. step length is reduced

$\Rightarrow$ overall stride length is also reduced

GRF remains posterior to the knee, producing an increased external knee flexor moment

Talipes equinus - forefoot fixed in plantarflexion
(spasticity in the triceps surae) plantarflexors
or weak dorsiflexors



Mild : foot may be placed on the floor

Severe - heel never contacts the ground

Toestrike

GRF line displaced anteriorly, inc. external knee extension moment

Loss of initial rocker

Stride length ↓

Foot deformities with excessive medial / lateral contact

Talipes equinovarus – curved foot where all the load is borne by the outer border of the forefoot (club foot)

Stamping : walking with forcible steps

Loss of sensation – person needs feedback on ground contact from the vibration / sound of impact

Insufficient push-off because forefoot cannot be loaded normally

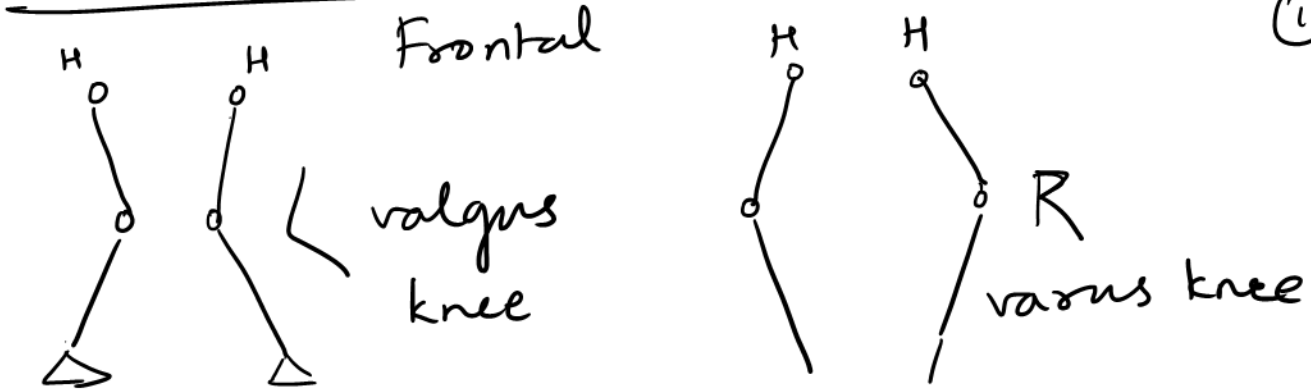
- problem w/ triceps surae
- problem w/ Achilles tendon
- pain in the forefoot

Asymmetry in gait timing

Shorten stance on affected leg

Shorten swing, ∴ shorter step length of opposite leg.

Abnormal walking base



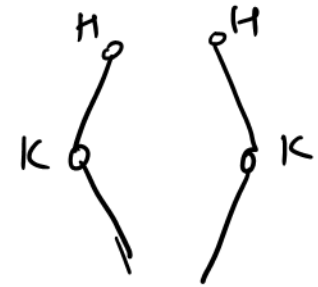
Larger walking base

① bcos of abducted hip / valgus knee → Lateral trunk bending

② bcos of instability & fear of falling – bcos sensation or proprioception is affected

Narrow base bcos of hip adduction deformity

or varus knee

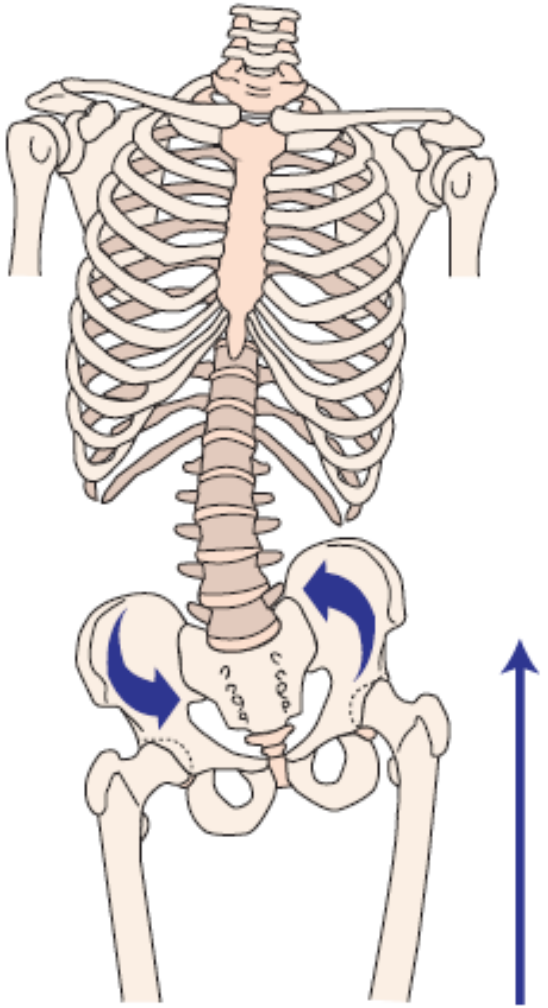


May cause the swinging leg to cross the midline

scissor gait

seen in cerebral palsy

Functional leg length discrepancy



Functional leg length discrepancy



Functional leg length discrepancy



Assistive devices for mobility

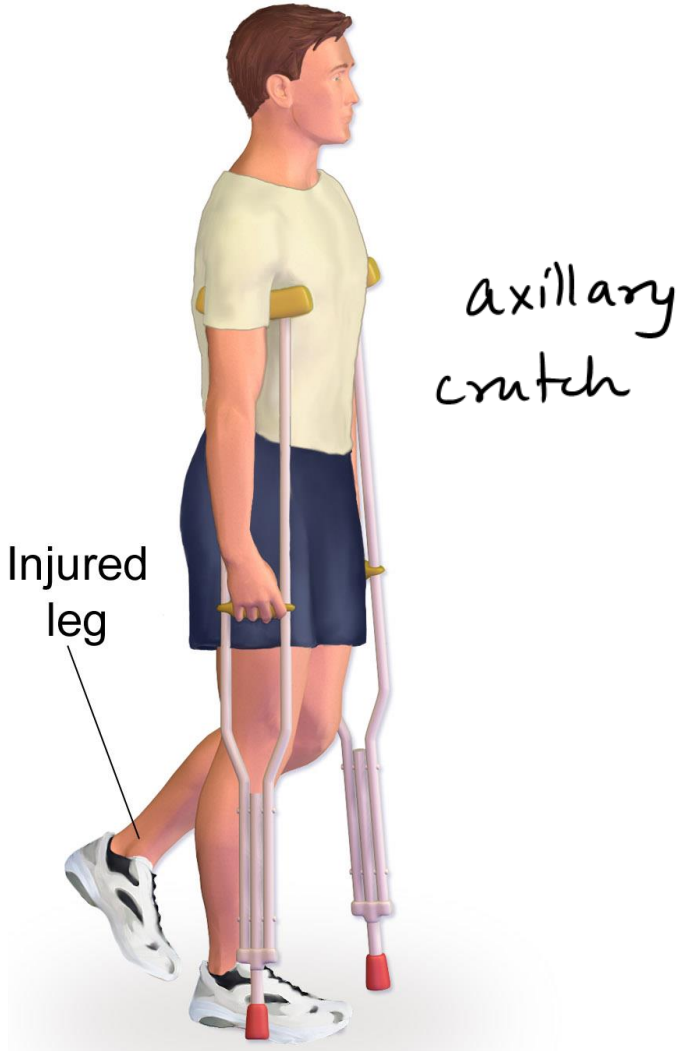


Forearm

Vissco Elbow/Forearm crutch



Ossenberg Elbow/Forearm crutch



Walking with Crutches

TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

By BruceBlaus - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=59441530